1. Let G be an abelian group. We define $D = \{(g, g) : g \in G\} \subseteq G \times G$. Use the First Isomorphism Theorem to prove that $(G \times G)/D \cong G$.

Consider the map $\varphi : G \times G \to G$ defined by $\varphi(a, b) = ab^{-1}$ for all $(a, b) \in G \times G$. Let $(a, b), (c, d) \in G \times G$. Then, using the fact that G is abelian,

$$\varphi((a, b)(c, d)) = \varphi(ac, bd) = (ac)(bd)^{-1} = acd^{-1}b^{-1} = (ab^{-1})(cd^{-1}) = \varphi(a, b)\varphi(c, d).$$

Thus φ is a homomorphism. Note that for any $g \in G$, we have $\varphi(g, 1) = g1^{-1} = g$. Thus φ is surjective, and hence an epimorphism. Notice that $(x, y) \in \ker \varphi$ if and only if $xy^{-1} = 1$. That is, $x = y$. Hence, $\ker \varphi = \{(g, g) : g \in G\} = D$. Then by the First Isomorphism Theorem, $(G \times G)/D \cong G$.

- Note that elements of $G \times G$ have the form (a, b) where $a, b \in G$. It is not necessarily true that $a = b$.

[15 marks]

2. (a) State the Second Isomorphism Theorem.

(b) Let G be a group of order pq, where p and q are distinct primes. Suppose G has subgroups H and K such that $|H| = p$, $|K| = q$ and $K \triangleleft G$. Prove that $HK = G$.

HINT: Find $|HK|$.

(a) **Second Isomorphism Theorem.** *Let H and K be subgroups of G with $K \triangleleft G$. Then $H/(H \cap K) \cong (HK)/K$.*

(b) By the Second Isomorphism Theorem, $HK/K \cong H/(H \cap K)$. Then $|HK/K| = |H/(H \cap K|$. Since these are finite groups, we can apply Lagrange's Theorem to get $|HK|/|K| = |H|/|H \cap K|$. That is, $|HK| = pq/|H \cap K|$. Note that $H \cap K$ is a subgroup of both H and K. By Lagrange's Theorem, $|H \cap K|$ must divide both p and q. However, p and q are distinct primes, so we must have $|H \cap K| = 1$. Thus $|HK| = pq/1 = pq = |G|$. Since HK is a subgroup of G with order equal to that of G, we must have $HK = G$.

[2 + 15 = 17 marks]

3. Let G be an abelian group (possibly infinite) with subgroups H and K such that $H \subseteq K$. Suppose $|G : K| = n$ and $|G : H| = m$ where $n \mid m$. Find $|K : H|$.

Since G is abelian, H and K are normal in G. We can apply the Third Isomorphism Theorem to get

$$(G/H)/(K/H) \cong G/K.$$

The orders of these two groups must then be equal. Note that $|G/H| = |G : H| = n$. So these are all finite groups. Applying Lagrange's Theorem, we get

$$|G/H|/|K/H| = |G/K|.$$

Then $|K/H| = |G/H|/|G/K| = |G : H|/|G : K| = m/n$. Thus $|K : H| = m/n$.

[15 marks]

4. Consider the group $G = \mathbb{Z}_{24}$ with subgroup $H = \langle [12] \rangle$.

(a) Determine all subgroups of $\mathbb{Z}_{24}$ and draw the lattice diagram.

(b) Use the Correspondence Theorem to draw the lattice diagram for $\mathbb{Z}_{24}/H$.

(a) By the Fundamental Theorem of Finite Cyclic Groups, the subgroups of $\mathbb{Z}_{24}$ correspond to the divisors of 24. Namely, 1, 2, 3, 4, 6, 8, 12 and 24. The subgroups of $\mathbb{Z}_{24}$ are the unique subgroups groups of these orders. Namely, $\{0\} = \langle 24 \rangle$, $\langle 12 \rangle$, $\langle 8 \rangle$, $\langle 6 \rangle$, $\langle 4 \rangle$, $\langle 3 \rangle$, $\langle 2 \rangle$, $\mathbb{Z}_{24} = \langle 1 \rangle$. The lattice diagram is:

$$\begin{tikzcd}
 & & & \mathbb{Z}_{24} = \langle 1 \rangle \arrow[dl, dash] \arrow[dr, dash] & \\
 & & \langle 2 \rangle \arrow[dl, dash] \arrow[dr, dash] & & \langle 3 \rangle \arrow[dl, dash] \\
 & \langle 4 \rangle \arrow[dl, dash] \arrow[dr, dash] & & \langle 6 \rangle \arrow[dl, dash] & \\
\langle 8 \rangle \arrow[dr, dash] & & H = \langle 12 \rangle \arrow[dl, dash] & & \\
 & \{0\} = \langle 24 \rangle & & &
\end{tikzcd}$$

(b)

$$\begin{tikzcd}
 & & \mathbb{Z}_{24}/H \arrow[dl, dash] \arrow[dr, dash] & \\
 & \langle 2 \rangle/H \arrow[dl, dash] \arrow[dr, dash] & & \langle 3 \rangle/H \arrow[dl, dash] \\
\langle 4 \rangle/H \arrow[dr, dash] & & \langle 6 \rangle/H \arrow[dl, dash] & \\
 & H/H & &
\end{tikzcd}$$

[10 + 5 = 15 marks]